

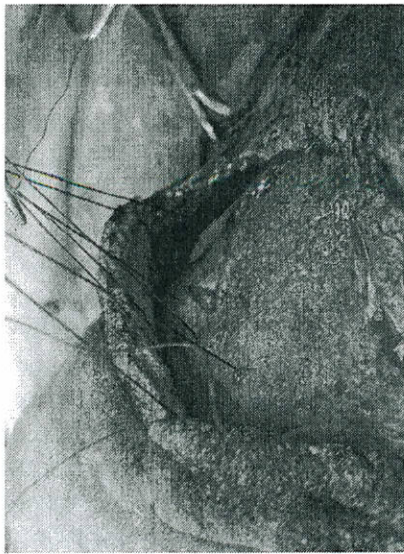


Figure 6: Walking sutures help to pull wound edges together and allow closure without excessive tension.

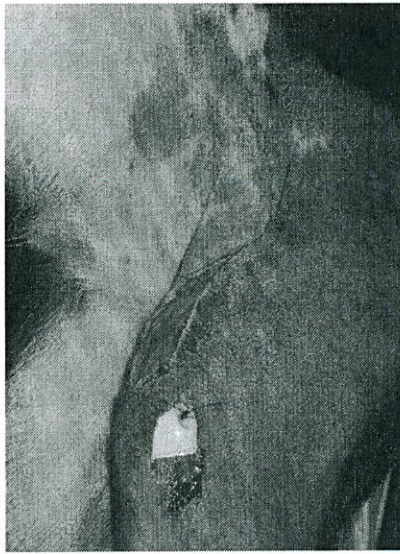


Figure 7: If there is doubt about infection or dead space, a drain should be placed. A penrose drain is shown. Note: This dog chewed on the incision and the wound opened. Re-suturing was necessary. Therefore placement of an E-collar is recommended.

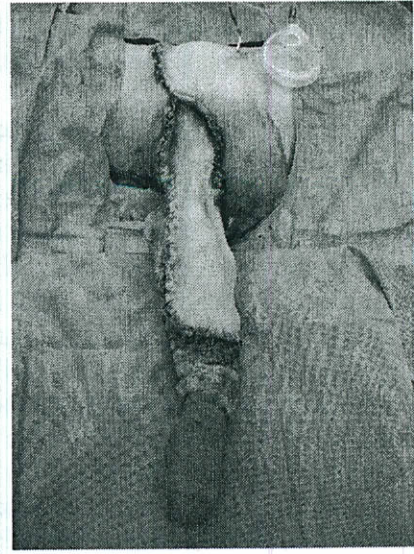


Figure 8: A closed suction drain provides drainage with minimal risk of wound contamination.

COMPLICATIONS WITH WOUND HEALING OR SKIN RECONSTRUCTION¹⁻⁵

Complications with skin reconstructive surgery include hemorrhage or hematomas, which are usually treated with the application of warm compresses. Seromas can form as well. In contrast to infected areas, seromas are not hot or inflamed and a complete blood count will not show signs of infection. Aspiration of seromas increases the risk of infection. Therefore, seromas should not be aspirated. The application of compression bandages and warm compresses will help with the resorption of the seroma. Dehiscence occurs usually three to five days after surgery. If there is too much tension, if there was poor tissue handling, or if there is too little blood supply to the area, dehiscence is likely. The cause for dehiscence should be determined and measures taken to prevent further negative effects on healing. For example, releasing incisions might need to be performed or open wound treatment initiated. Infections of surgical wounds occur in 2.5% of clean surgeries⁵, and may be prevented by the use of strictly aseptic techniques and intraoperative antibiotics. A common protocol is the administration of Cephazolin (22mg/kg IV) at induction followed by intravenous administration every two hours during surgery. Signs of infection usually occur two to three days after surgery. Common cultured bacteria from surgical wounds include: *Staphylococcus species*, *E.coli*, *Bacteroides*, *Clostridium*, and *Pasteurella*. When facing an infection, culture and sensitivity tests should be initiated and surgical drains should be placed. A broad-spectrum antibiotic should be administered until culture and sensitivity tests reveal the most potent and appropriate antimicrobial agent to be used. Finally, problems like delayed- or incomplete wound healing and wound contracture may need to be addressed. If an atypical wound is present, or if the wound is not responding as expected, the following causes have to be considered: Mycobacterial infection, bacterial or fungal disease, underlying host factors (Cushing's disease, diabetes, renal disease etc.), underlying osteomyelitis, or polycythemia. Depending on the cause, medical therapy, tension-relieving techniques, skin flaps or free skin grafts may need to be performed in order to address these problems.

UNCONVENTIONAL WOUND TREATMENT METHODS

Unconventional wound treatment methods include the use of honey, sugar, maggots and leeches. These measures are not adequate for wound treatment on the trail, but may need to be applied occasionally under the guidance of a veterinarian to treat specific wound of dogs in the kennel. Medical grade honey has been shown to have bactericidal effect, and appears in some cases superior to many modern methods of wound treatment. Wounds treated with honey show decreased inflammation and accelerated sloughing of necrotic tissue. Honey provides a local cellular energy source, forms a

protective layer of protein over the wound, and forms a healthy granulation bed. Large case studies in veterinary medicine are lacking, and despite amazing results in the human field the use of honey has been slow to come into acceptance in our profession.⁶

Granulated sugar has also been proven to be beneficial in the treatment of open, contaminated wounds. Sugar reduces the need for debridement because macrophages are drawn into the wound, which cleans the wound and accelerates sloughing of devitalized tissue. Its use enhances healing and eliminates the need for antibiotics because of its rapid antibacterial action. As with honey, its use in animals is not yet widespread.⁷

Maggot debridement therapy is the medical use of specially selected and tested, disinfected fly larvae ("maggots") for cleaning non-healing wounds. In 2004, the Food-and-Drug-Administration (FDA) approved the marketing of medical maggots. Medicinal maggots have been found to have four principle actions: 1) They debride (clean) wounds by dissolving the necrotic, infected tissue. 2) They disinfect the wound by killing bacteria. 3) They stimulate wound healing. 4) They break down and inhibit the formation of biofilm. Maggot therapy was used in the early 20th century until it was replaced by the use of antibiotics. With the increasing amount of antibiotic resistance, this form of therapy for non-healing, infected wounds, is returning. It is uncommonly used in animals, but studies in humans show that this form of wound treatment may help in cases, in which conventional wound treatment methods are unsuccessful.⁸ Leeches can be used to relieve severe venous congestion, following skin-flap procedures. While some reports support their use, other authors raise concerns about the increased risk of infection, or the efficacy of leeches.⁸ It is important to understand that these wound treatment techniques have to be monitored closely by a veterinarian and should not be applied without specific knowledge of wound treatment.

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Footnotes:

a) Stryker Interpulse® System, Kalamozoo, MI 49001, <http://www.stryker.com>. b) The Kendall Company, Mansfield, MA 02048, <http://www.kendallhq.com>. C) Closed "Blake Drain kit": www.ethicon.com.

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